1.

Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.

ANS:-

DNS Resolution Failure:

- Steps:
    - Edit the hosts file and point www.spotify.com to a wrong IP address, such as 192.0.2.1.
    - Open Command Prompt and run: ping www.spotify.com
    - Observe the response.

Error/Result: Request timed out.

Explanation: The wrong IP address in the hosts file redirects www.spotify.com to an incorrect destination, so the ping request cannot reach the actual Spotify server.

2.

Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

ANS:-

DHCP Incorrect Default Gateway:

- Configuration:
    - DHCP Server: Configured with an incorrect default gateway, e.g. 192.168.1.99
    - Client: Connected to the DHCP server and set to DHCP.

- Verification: The client received the incorrect gateway 192.168.1.99 automatically from the DHCP server.
- How to Identify and Fix:
    - Check the client's IP Configuration and verify the default gateway.
    - Compare it with the actual router/gateway address.
    - Correct the default gateway in the DHCP server settings and renew the client's IP configuration.

Result: After correction, the client receives the correct gateway and can communicate with other networks.

3.

Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

ANS:-

DNS Troubleshooting:

- Troubleshooting Steps:
    - Run: nslookup www.google.com to check whether the DNS server is resolving the website name.
    - Check the DNS server address configured on the client device.
    - Test the DNS server again using nslookup and check for a valid IP address.
    - If the DNS server is incorrect or unavailable, configure the correct DNS server and test again.

- Solution: The problem was caused by an incorrect/unavailable DNS server. After configuring the correct DNS server, website names resolved properly and the devices could access websites normally.

4.

A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it. Hint: Check for network connectivity, DHCP scope exhaustion, and server availability.

ANS:-

DHCP Troubleshooting:

- Possible Causes:
    - Network connection between the device and DHCP server is down.
    - DHCP scope has no available IP addresses.
    - DHCP server is stopped or unavailable.
    - Incorrect network/VLAN configuration.
- Troubleshooting Steps:
    - Check network connectivity between the device and DHCP server.
    - Check the DHCP scope to see if all IP addresses are already used.
    - Check whether the DHCP server is running and reachable.
    - If the scope is full, increase the DHCP address range or free unused addresses.
    - Renew the client's IP address and check whether it receives a valid IP.

Result: The APIPA address 169.x.x.x appears when the device cannot get an IP address from the DHCP server. After fixing the DHCP/network issue, the device should receive a valid IP address automatically.

5.

Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

ANS:-

Common DNS and DHCP Issues:

N+- Troubleshooting Network Services.pdf

Type	Issue	Symptoms	Possible Cause	Recommended Solution
DNS N+- Troubleshooting Network Services.pdf	DNS Server Failure N+- Troubleshooting Network Services.pdf	Websites do not open by name N+- Troubleshooting Network Services.pdf	DNS server is unavailable N+- Troubleshooting Network Services.pdf	Check and configure the correct DNS server N+- Troubleshooting Network Services.pdf
DNS N+- Troubleshooting Network Services.pdf	Wrong DNS Record N+- Troubleshooting Network Services.pdf	Domain opens incorrectly N+- Troubleshooting Network Services.pdf	Incorrect A/CNAME record N+- Troubleshooting Network Services.pdf	Correct the DNS record N+- Troubleshooting Network Services.pdf
DNS N+- Troubleshooting Network Services.pdf	DNS Cache Issue N+- Troubleshooting Network Services.pdf	Old or incorrect IP is returned N+- Troubleshooting Network Services.pdf	Outdated DNS cache N+- Troubleshooting Network Services.pdf	Clear the DNS cache and try again N+- Troubleshooting Network Services.pdf
DHCP N+- Troubleshooting Network Services.pdf	DHCP Server Down N+- Troubleshooting Network Services.pdf	Device gets $169.x.x.\times IP$ N+- Troubleshooting Network Services.pdf	DHCP service is not running N+- Troubleshooting Network Services.pdf	Restart/check the DHCP server N+- Troubleshooting Network Services.pdf
DHCP N+- Troubleshooting Network Services.pdf	DHCP Scope Exhausted N+- Troubleshooting Network Services.pdf	New device cannot get an IP N+- Troubleshooting Network Services.pdf	All IP addresses are already used N+- Troubleshooting Network Services.pdf	Increase the DHCP pool or free unused IPs N+- Troubleshooting Network Services.pdf

Type	Issue	Symptoms	Possible Cause	Recommended Solution
DHCP N+- Troubleshooting Network Services.pdf	Wrong DHCP Settings N+- Troubleshooting Network Services.pdf	Device gets incorrect gateway/IP N+- Troubleshooting Network Services.pdf	Incorrect DHCP configuration N+- Troubleshooting Network Services.pdf	Correct the DHCP scope settings N+- Troubleshooting Network Services.pdf